

CODEALPHA BIOINFORMATICS INTERNSHIP

TASK-2

Title-Perform Multiple sequence Alignment (MSA) assignment.

Protein Family Selected: Cytochrome c oxidase subunit I (COX1) family

(Highly conserved mitochondrial protein; commonly used for sequence comparison)

1. Protein sequences selected (5 members)

FASTA sequences from **NCBI/UniProt**:

Homo sapiens COX1

>Human_COX1

```
MFADRWLFSTNHKDIGTLYLLFGAWAGVLGTALSLIRAELGQPGNLLGNDHIYNVIVTA
HAFVMIFFMVMPIMIGGFGNWLVLPLMIGAPDMAFPRMNNMSFWLLPPSLLLLASAMVEA
GAGTGWTVYPPLAGNYSHPGASVDLTIFSLHLAGVSSILGAINFITTIINMKPPAMTQYQ
TPLFVWSVLITAVLLLLSLPVLAAGITMLLTDRNLNTTFFDPAGGGDPILYQHLFWFFGH
PEVYILILPGFGMISHIVTYYS GKKEPFGYMGMVWAMMSIGFLGFIVWAHMHMFTVGMDVD
TRAYFTSATMIIAIP TGVKVFSWLATLHGSNMKWSAAVLWALGFIFLFTVGGLTGIVLAN
SSLDIVLHDTYYVVAHFHYVLSMGAVFAIMGGFIHWFP LFSGYTLDQTYAKIHFTIMFIG
VNLTFFPQHFLGLSGMPRRYS DYPDAYTTWNILSSVGSFISLTAVMLMIFMIWEAFASKR
KVL MVEEPSMNLEWLYGCPPPYHTFEPPVYMKS
```

Mus musculus COX1

>Mouse_COX1

MFINRWLFSTNHKDIGTLYLLFGAWAGMVGTA
LSILIRAE LGQPGALLGDDQIYNVIVTA
HAFVMIFFMVMPMMIGGFGNWLVP
LMIGAPDMAFPRMNNMSFWLLPPSFL
LLLASSMVEA
GAGTGWTVYPPLAGNLAHAGASVDLT
IFSLHLAGVSSILGAINFITTIINMKPP
AMTQYQ
TPLFVWSVLITAVLLLLSLPVLAA
GITMLLTDRNLNTTFFDPAGGGDPILY
QHFWFFGH
PEVYILILPGFGIISHVVITYSGKKEP
FGYMGMVWAMMSIGFLGFIVWAHHMFT
VGLDVD
TRAYFTSATMIIAIP
TGKVF
SWLATLHGGNIKWSPAM
LWALGFIFLFTVGGLTGIVLSN
SSLDIVLHDTYYVVAHFHYVLSMGAV
FAIMAGFVHWFLFSGFTLDDTWAKAH
FAIMFVG
VNMTFFPQHFLGLSGMPRRYSDY
PDAYTTWNTVSSMG
SFISLTAVLIMIFMIWEAFASKR
EVMSVSYASTNLEWLHGCPPPYHT
FEEPTYVKVK

Bos taurus COX1

>Cow_COX1

MFINRWLFSTNHKDIGTLYLLFGAWAGMVGTA
LSILIRAE LGQPGTLLGDDQIYNVIVTA
HAFVMIFFMVMPIMIGGFGNWLVP
LMIGAPDMAFPRMNNMSFWLLPPSFL
LLLASSMIEA
GAGTGWTVYPPLAGNLAHGGASVDLT
IFSLHLAGVSSILGAINFITTIINMKPP
AMTQYQ
TPLFVWSVLVTAVLLLLSLPVLAA
GITMLLTDRNLNTTFFDPAGGGDPILY
QHFWFFGH
PEVYILILPGFGMISHIATYSGKKEP
FGYMGMVWAMMSIGFLGFIVWAHHMFT
VGMDVD
TRAYFTSATMIIAIP
TGKVF
SWLATLHGGNIKWSPALLWALGFIFLFT
VGGLTGIVLAN
SSLDVVLHDTYYVVAHFHYVLSMGAV
FAIMGGFVHWFLFSGYTLNQTWAKIH
FLIMFVG
VNLTFFPQHFLGLSGMPRRYSDY
PDAYTMWNTVSSMG
SFISLTAVILMVFMVWEAFASKR
EVSTVELTTTNLEWLNGCPSYHT
FEETPAYVKI

Danio rerio COX1

>zebrafish_COX1

MTITRWWFFSTNHKDIGTLYLVFGAWAGMVGTA
LSLLIRAE LSQPGALLGDDQIYNVIVTA

HAFVMIFFMVMPIIGGFGNWLVLPLMIGAPDMAFPRMNNMSFWLLPPSFLLLASSGVEA
GAGTGWTVYPPLAGNLAHAGASVDLTIFSLHLAGVSSILGAINFITTTINMKPPTISQYQ
TPLFVWAVLVTAVLLLLSLPVLAAGITMLLTDRNLNTTFFDPAGGGDPILYQHLEWFFGH
PEVYILILPGFGIISHVVAYYAGKKEPFGYMGVMWAMMAIGLLGFIVWAHHMFTVGMDVD
TRAYFTSATMIIAIPGKVFVSWLATLHGGAIKWETPMLWALGFIFLFTVGGLTGIVLAN
SSLDIVLHDTYYVVAHFHYVLSMGAVFAIMAGFVHWPLFTGYTLNSVWTKIHFGVMFIG
VNLTFFPQHFLGLAGMPRRYSYDYPDAYALWNTVSSIGSLSLVAVIMFLFILWEAFTAKR
EVLSVELTATNVEWLHGCPPPYHTFEPAFVQIQSN

Saccharomyces cerevisiae COX1

>Yeast_COX1

MVQRWLYSTNAKDIAVLYFMLAIFSGMAGTAMSLIIRLELAAPGSQYLHGNSQLFNVLVV
GHAVLMIFFLVMPALIGGFGNYLLPLMIGATDTAFPRINNIAFWVLPMLVCLVTSTLVE
SGAGTGWTVYPPLSSIQAHSGPSVDLAIFALHLTSISSLLGAINFIVTTLNMRTNGMTMH
KLPLFVWSIFITAFLLLLSLPVLSAGITMLLLDRNFNTSFFEVSAGGDPILYEHLEWFFG
HPEVYILIIPGFGIISHVVSTYSKKPVFGEISMVYAMASIGLLGFLVWSHHMYIVGLDAD
TRAYFTSATMIIAIPGKIFSWLATIHGGSIRLATPMLYAI AFLFTMGGLTGVALAN
ASLDVAFHDTYYVVGHFHYVLSMGAI FSLFAGYYYWSPQILGLNYNEKLAQIQFWLIFIG
ANVIFPMMHFLGINGMPRRIPDYPDAFAGWNYVASIGSFIATLSLFLFIYILDQLVNGL
NNKVNNKSVIYNKAPDFVESNTIFNLNTVKSSSIEFLTSPPAVHSFNTPAVQS

2. Multiple Sequence Alignment

Tool used: **Clustal Omega**

Website:

<https://www.ebi.ac.uk/jdispatcher/msa/clustalo>

Input:

Upload the 5 FASTA protein sequences

Run alignment

Download alignment in **Clustal format**

Conserved Regions and Motifs Identified

The screenshot displays the Clustal Omega web interface. At the top, there is a navigation bar with links for Home, Help & Privacy, Recent Jobs, and Input Form. Below this, a banner for Clustal Omega Multiple Sequence Alignment (MSA) is visible. A yellow notification bar mentions the UniProtKB release 2026_02. The main content area describes Clustal Omega as a new multiple sequence alignment program. The 'Input sequence' section includes a 'Sequence type' dropdown set to 'Protein', a text area for pasting sequences (containing an example of Human COX1), and a 'Choose File' button. At the bottom, there are buttons for 'Use the example', 'Clear sequence', and 'More example inputs'.

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Clustal Omega

Multiple Sequence Alignment (MSA)

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UniProtKB release 2026_02 is now available. This release includes a new dataset [UniProt Pan Proteomes](https://insideuniport.blogspot.com/2026/06/capturing-diversity-of-life.html), which contains selected proteins from proteomes in UniProtKB and UniParc. UniProtKB Reference Proteomes plus Swiss-Prot and UniProtKB Reference Proteomes datasets have been retired. For more information please refer to <https://insideuniport.blogspot.com/2026/06/capturing-diversity-of-life.html>

Clustal Omega is a new multiple sequence alignment program that uses seeded guide trees and HMM profile-profile techniques to generate alignments between three or more sequences. For the alignment of two sequences please instead use our pairwise sequence alignment tools. This tool can align up to 4000 sequences or a maximum file size of 4 MB.

Input sequence

Sequence type

☒ Protein ☐ DNA ☐ RNA

Paste your sequence here - or use the example sequence

```
>Human_COX1
MFADRWLFSTNHKDIGTLYLLFGAWAGVLGTALSLIRAE LGQPGLGNDHIYNVIVTA
HAFVMIFFMVMPIIMIGGFGNWLPLMIGAPDMAFPRMNMNSFWLLPSSL LLLASAMVEA
GAGTGWT VYPLAGNYSHPGASVDLTIFSLHLAGVSSILGAINFIT IINMKPPAMTQYQ
TPLFVWSVLITAVLLLSLPVLAAGITMLLDRNLNTTFFDPAGGGDPILYQH LFWFFGH
PEVYLILPGFGMISHIVTYYSKGKPEFGYMGVWAMMSIGFLGFIVWAHMHMFTVGMDVD
TRAYFTSATMIIAIP TGKVFSLATLHGSNMKWSAAVLWALGFIFLFTVGGLTGIVLAN
```

Choose File COX1_4species.fasta.docx **Use the example** **Clear sequence** More example inputs

Parameters **OUTPUT FORMAT**

Conserved catalytic motifs

A highly conserved COX1 region involved in electron transport:
and conserved histidine residues:

Participate in binding heme groups

Essential for oxygen reduction activity

Interpretation of MSA Results

High sequence similarity

All five COX1 proteins show strong conservation. The large number of identical residues indicates evolutionary conservation.

Functional regions are conserved

Hydrophobic membrane-spanning regions remain unchanged because they are required for mitochondrial membrane localization.

Conserved amino acids indicate functional importance

Histidine residues and other conserved motifs are maintained because they participate in enzyme activity.

Sequence variation

Differences occur mostly in less functionally important regions.

These variations reflect evolutionary divergence between organisms.

Conclusion

The MSA of five COX1 protein sequences using Clustal Omega demonstrates that members of the same protein family share highly conserved amino acid regions. Conserved motifs and catalytic residues are maintained across species, supporting their essential role in mitochondrial electron transport and energy production.

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